Sharing Data: BigQuery Sharing and Clean Rooms

Copies, authorised views and listings; exchanges, listings and linked datasets; who pays; egress controls; data clean rooms and analysis rules.

SECTION 4 of 5 EXAM WEIGHT 15% MODULE 32 of 45

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Share the live data, not a copy. Then decide what the other side may do with it.

Four ways to share, compared

OPTION	BEST WHEN	COPIES DATA?
Export or copy	A one-off file for a system outside Google Cloud	Yes, and it goes stale
Authorised view	Teams inside your own organisation	No
BigQuery sharing listing	Other organisations, discoverable and live	No: a linked dataset
Data clean room	Joining data with a partner, neither sees rows	No

THE TIEBREAKER

Another organisation, live data, no copies: a listing. Combining both sides' data privately: a clean room.

Exchanges, listings and linked datasets

- An **exchange** groups listings. Private by default, or public to every Google Cloud user.
- A **listing** points at a dataset or a Pub/Sub topic, with docs and sample queries.
- Subscribing creates a **linked dataset**: read-only, live, and nothing is copied.
- Analytics Hub is now called **BigQuery sharing**. Roles still say Analytics Hub.

> THE NUMBERS
>
> One shared dataset can feed up to 1,000 linked datasets, each in a subscriber's own project.

What can be shared

- Datasets: tables, views, authorised views, materialized views, routines and ML models.
- **Pub/Sub topics**, for live streaming data, as linked subscriptions.
- Row-level security and masking work, but subscribers need **Enterprise** or Enterprise Plus.
- Public listings can be **sold** through Cloud Marketplace.

> THE TRAP
>
> Shared view definitions show in query plans. Never put keys or secrets in them.

Who pays for what?

PUBLISHER

Pays to keep it

- Storage of the shared data
- Pub/Sub publishing, for topics
- No charge for exchanges or listings

SUBSCRIBER

Pays to use it

- Every query on the linked dataset
- On-demand bytes, or their own slots
- Pub/Sub subscribing, for topics

THE SIGNAL

The partner's queries are billed to the partner. That is often the deciding phrase.

Egress controls and limits

- **Egress controls** on a listing stop copy, clone, export, snapshot and transfers.
- They can also block **CREATE TABLE AS SELECT** and writing results to tables.
- No materialized views or snapshots over linked tables. Huge joins may fail.
- The listing's dataset must be in the **same region** as the exchange.

> REMEMBER
>
> The publisher keeps the encryption key. Disable it, and every subscriber loses access too.

Sharing regional demand with Westgrid

01
Publish
marts.region_demand_hh, in europe-west2.

02
Exchange
Private exchange lk-partners.

03
Listing
Egress restricted, docs, sample queries.

04
Subscribe
Westgrid gets a live linked dataset.

05
Query
In Westgrid's project, on Westgrid's bill.

THE PAYOFF

Updates land the moment Larkspur writes them, so the five-minute target holds.

Joining data nobody may see

- Built on sharing. An **owner** runs it, **contributors** add data, **subscribers** query.
- Contributors set **analysis rules** on views, so raw rows never come back.
- **Query templates**, built from table functions, allow only approved queries.
- Egress of raw data is blocked automatically. Results can be blocked too.

> HONEST LIMIT
>
> Google says rules are not guaranteed to block every extraction attempt. Combine them with templates.

Four analysis rules

RULE	WHAT IT ALLOWS	QUERY SHAPE
Aggregation threshold	Totals only, over at least N people per row	SELECT WITH AGGREGATION_THRESHOLD
Differential privacy	Noisy totals, within a privacy budget	SELECT WITH DIFFERENTIAL_PRIVACY
List overlap	Only rows that match after a join	Join on the allowed columns
Join restriction	Joins only on chosen columns, or none	JOIN_ALL, JOIN_ANY, JOIN_BLOCKED

THE TRAP

When the privacy budget is spent, the view cannot be queried until it is updated or re-created.

Larkspur's side of the Hearthly clean room

```
1  CREATE OR REPLACE VIEW clean.lk_households
2  OPTIONS (privacy_policy = '''{"aggregation_threshold_policy":
3    {"threshold": 50, "privacy_unit_column": "household_key"}}''')
4  AS SELECT household_key, region, tariff_type, avg_daily_kwh
5  FROM shared.hearthly_ready;
6
7  -- What Hearthly may run: totals only, 50 households or more per row
8  SELECT WITH AGGREGATION_THRESHOLD region, tariff_type, COUNT(*) AS homes
9  FROM clean.lk_households
10 GROUP BY region, tariff_type;
```

2-3 Every result row must cover at least 50 distinct households.

8 Queries must aggregate. Rows under the threshold are dropped.

REMEMBER

The household key can never be selected directly, only counted.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
Another **organisation**, live, no copies	BigQuery sharing listing
Partner pays **their own queries**	Listing, linked dataset in their project
They must **not export** or copy it	Egress controls on the listing
Combine lists, **neither sees the other's**	Data clean room
Totals only, never single people	Aggregation threshold rule
Queries fail after **many runs** on a DP view	Privacy budget used up

THE PATTERN

Who is it for, who pays, and what may they do with it: three questions, three features.

> Westgrid wants Larkspur's live half-hourly demand data inside its own Google Cloud organisation.
>
> Larkspur must not pay for Westgrid's queries, and Westgrid must not be able to copy or export the data.

What should Larkspur set up?

A Export the table as CSV files to a shared bucket every hour

B A private listing with egress restricted, which Westgrid subscribes to

C An authorised view, with Westgrid's users running jobs in Larkspur's project

D A nightly DTS copy of the table into a dataset in Westgrid's project

◆ PAUSE AND DECIDE

What should Larkspur set up?

A Hourly CSV export

Copies that are stale and easily passed on.

B Private listing

Live, billed to Westgrid, and exports are blocked.

C Authorised view

Queries run in Larkspur's project, on Larkspur's bill.

D Nightly copy

A day out of date, and the copy is theirs to export.

THE SIGNAL

Another organisation, live, partner pays, no exports: a listing with egress controls.

> Larkspur and Hearthly want to know how many customers they share, by region.
>
> Neither company may see the other's customer list, or any individual customer's details.

What should the two companies use?

A Swap lists of hashed email addresses by SFTP, then compare them

B A linked dataset of Larkspur's customers, shared with Hearthly

C An authorised view that joins both tables, shared with both sides

D A data clean room, with list overlap or threshold rules

◆ PAUSE AND DECIDE

What should the two companies use?

A **Swap hashed lists**
Each side holds the other's list, and hashes can be guessed.

B **Linked dataset**
Hearthly sees Larkspur's whole customer list.

C **Authorised view**
Someone must hold both tables, and rows come back.

D **Clean room**
Matching happens inside, and only totals come out.

THE SIGNAL

Combine two parties' data, neither sees rows: a data clean room.

> Westgrid subscribes to a Larkspur listing whose tables carry row access policies. Its queries on the linked dataset fail.
>
> Westgrid runs BigQuery on the Standard edition.

What is the most likely fix?

A Westgrid moves to Enterprise or Enterprise Plus edition

B Larkspur grants Westgrid Fine-Grained Reader on the tables

C Larkspur removes the egress controls from the listing

D Larkspur moves the exchange into the US multi-region

◆ PAUSE AND DECIDE

What is the most likely fix?

A Higher edition
Required to query listings with row-level security.

B Fine-Grained Reader
A column role. It does not fix the edition.

C Remove egress controls
Controls exports, not whether queries run.

D Move the exchange
Must stay with the dataset, in Europe.

THE SIGNAL

Shared tables with row-level security: subscribers need Enterprise or Enterprise Plus.

> Hearthly's analysts ran hundreds of queries against a differential privacy view in the clean room.
>
> Now every query against that view fails, although nothing else has changed.

What has most likely happened?

A The linked dataset expired after a fixed number of days

B An egress control has started blocking all of their queries

C The privacy budget is spent, so the view must be updated

D The aggregation threshold was raised above the data size

◆ PAUSE AND DECIDE

What has most likely happened?

A Linked dataset expired

Linked datasets do not expire on their own.

B Egress control

Blocks exports, not ordinary aggregate queries.

C Budget spent

Each query uses epsilon. At zero, the view is locked.

D Threshold raised

Nothing changed, and this is a different rule.

THE SIGNAL

Many queries, then a differential privacy view stops working: the budget is spent.

Module 32 in five lines

- Share live data, not copies. **Authorised views** inside, **listings** across organisations.
- A listing gives a read-only **linked dataset**. The subscriber pays for queries.
- **Egress controls** stop copying and exporting. Row security needs Enterprise.
- **Clean rooms** combine partners' data, and only aggregates come out.
- Rules: **aggregation threshold**, differential privacy, list overlap, join restriction.

COMING UP

Next: module 33, the start of section five: BigQuery capacity and pricing.